# Acceleration, signs and turning around

**LEARNING TARGET · *R*4** I can find acceleration from changes in velocity.

## A · KEY VOCABULARY

| | |
|---|---|
| **Average $a$** | Change in velocity divided by elapsed time. |
| **Speeding up** | $v$ and $a$ have the same direction in 1D. |
| **Slowing down** | $v$ and $a$ oppose each other before the turn. |
| **Turn** | $v = 0$ for an instant; $a$ can be nonzero. |

## B · KEY EQUATIONS + UNITS

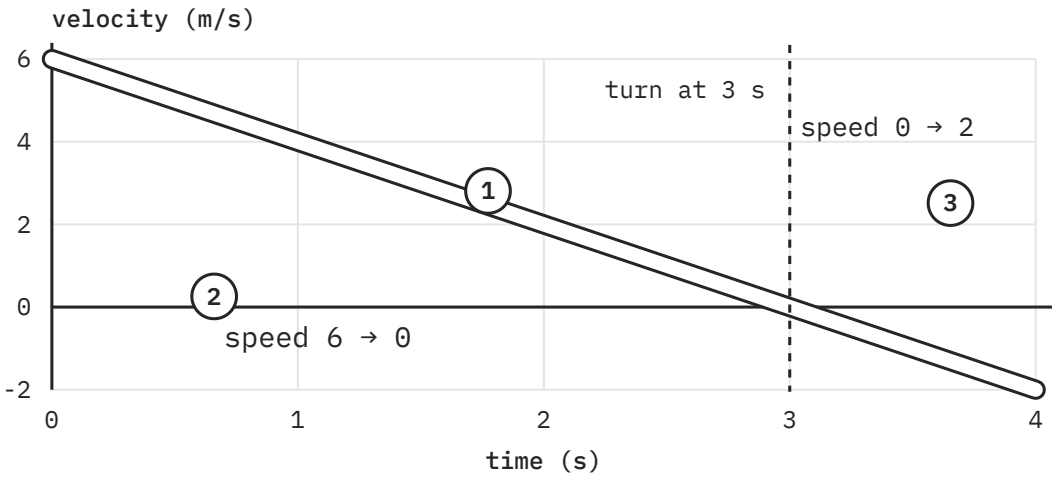
$$\bar{a} = \Delta v \div \Delta t$$

$$v_f = v_i + at$$

| | |
|---|---|
| $a$ | m/s² |
| $v$ `rule` | constant $a$ only |

## C · COLOR IT · SLOW, THEN SPEED UP

*Right is +; constant a.*




☐ **1 · PURPLE** Signed velocity

☐ **2 · RED** Before turn: slowing

☐ **3 · GREEN** After turn: speeding up

## D · WORKED EXAMPLE

A cart's velocity changes uniformly from **+6 m/s** to **−2 m/s** in **4 s**. **Find its acceleration and the time it reverses.**

| GIVEN | EQUATION | WORK | ANSWER |
|---|---|---|---|
| $v_i = +\mathbf{6}$ **m/s**<br>$v_f = -\mathbf{2}$ **m/s**<br>$\Delta t = \mathbf{4}$ **s** | $a = \Delta v/\Delta t$.<br>Turn: $0 = v_i + at$. | $a = (-2 - 6)/4$<br>$0 = 6 - 2t$<br>$t = \mathbf{3}$ **s** | $a = -\mathbf{2}$ **m/s²**<br>Slows for **3 s**, then speeds up to the left. |

## E · THE 2026-XJ THREAD

DAY 7 OF 22 · NEW TOOL: SIGNED CHANGE

A change in XJ's velocity requires an acceleration term in the forecast. Its sign alone does not show whether the asteroid is speeding up; compare the directions of velocity and acceleration.

*Unit question: How fast is it moving, and where will it arrive?*

**UNIT 1 COLOR CODE** ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else